

```
1
2 // COS30008 - Midterm 2026
3
4 #pragma once
5
6 #include <cctype>
7
8 template<size_t N>
9 class KeyIterator
10 {
11 public:
12
13     using difference_type = std::ptrdiff_t;
14     using value_type = char;
15     using iterator = KeyIterator;
16
17     KeyIterator(const char aInit[N]) noexcept
18     {
19         for (size_t i = 0; i < N + 1; i++)
20         {
21             fKeys[i] = std::isalpha(aInit[i] - 'A');
22         }
23         fIndex = 0;
24         fUpdateIndex = N;
25     }
26
27     size_t operator*() const noexcept
28     {
29         return fKeys[fIndex];
30     }
31
32     iterator& operator++() noexcept
33     {
34
35         fIndex++;
36         if (fIndex > N + 1) {
37             fIndex = 0;
38         }
39
40         return *this;
41     }
42
43     iterator operator++(int) noexcept
44     {
45         iterator lChange = *this;
46         (*this)++;
47         return lChange;
48     }
49 }
```

```
50     }
51
52     iterator& operator+=(char aKey) noexcept
53     {
54         fKeys[fUpdateIndex] = std::toupper(aKey - 'A');
55         fUpdateIndex++;
56         if (fUpdateIndex >= N + 1) {
57             fUpdateIndex = 0;
58         }
59
60         return *this;
61     }
62
63 private:
64     char fKeys[N+1];
65     size_t fIndex;
66     size_t fUpdateIndex;
67 };
68
69
```